

Solve each system of equations.

1.

$$5x - y = -7z + 16$$

$$x + y - 2z = -5$$

$$-2y + 14z = -10x + 8$$

$$5x - y + 7z = 16$$

$$x + y - 2z = -5$$

$$10x - 2y + 14z = 8$$

$$\begin{bmatrix} 5 & -1 & 7 & | & 16 \\ 1 & 1 & -2 & | & -5 \\ 10 & -2 & 14 & | & 8 \end{bmatrix}$$

No sol.

2.

$$x + 6z = 5$$

$$3x + y - 4z = -3$$

$$2x - y - 2z = -8$$

$$\begin{bmatrix} 1 & 0 & 6 & | & 5 \\ 3 & 1 & -4 & | & -3 \\ 2 & -1 & -2 & | & -8 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & | & -1 \\ 0 & 1 & 0 & | & 4 \\ 0 & 0 & 1 & | & 1 \end{bmatrix} = (-1, 4, 1)$$

3.

$$-2z = 9 - x - y$$

$$4x - y - 2z = 8$$

$$4x + 4z = -y - 6$$

$$\begin{array}{l} x + y - 2z = 9 \\ 4x - y - 2z = 8 \\ 4x + y + 4z = -6 \end{array} \Rightarrow \begin{bmatrix} 1 & 1 & -2 & | & 9 \\ 4 & -1 & -2 & | & 8 \\ 4 & 1 & 4 & | & -6 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & | & 1 \\ 0 & 1 & 0 & | & 2 \\ 0 & 0 & 1 & | & -3 \end{bmatrix} = (1, 2, -3)$$

4.

$$x + y + z = 8$$

$$3x - y + 4z = 25$$

$$6x - 2y + 8z = 50$$

$$\begin{bmatrix} 1 & 1 & 1 & | & 8 \\ 3 & -1 & 4 & | & 25 \\ 6 & -2 & 8 & | & 50 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1.25 & | & 8.25 \\ 0 & 1 & -.25 & | & -.25 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \text{ inf. solutions}$$

5.

$$4x - y - z = 27$$

$$x - y + 2z = -3$$

$$2x + y - 2z = 21$$

$$\begin{bmatrix} 4 & -1 & -1 & | & 27 \\ 1 & -1 & 2 & | & -3 \\ 2 & 1 & -2 & | & 21 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & | & 4 \\ 0 & 1 & 0 & | & 1 \\ 0 & 0 & 1 & | & -4 \end{bmatrix} = (6, 1, -4)$$

6.

$$x + 4y + 3z = -3$$

$$x = 4y + z + 9$$

$$2x + 4y = -z - 3$$

$$\begin{array}{l} x + 4y + 3z = -3 \\ x - 4y - z = 9 \\ 2x + 4y + z = -3 \end{array} \Rightarrow \begin{bmatrix} 1 & 4 & 3 & | & -3 \\ 1 & -4 & -1 & | & 9 \\ 2 & 4 & 1 & | & -3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & | & 2 \\ 0 & 1 & 0 & | & -2 \\ 0 & 0 & 1 & | & 1 \end{bmatrix} = (2, -2, 1)$$

For each problem (7-12), write a system of equations, solve the system, write your answers in context.

7. The Garza family enrolled their 4-year-old, 3-year-old, and infant in daycare for \$2800 per month. The Nguyen family enrolled their 4-year-old and 3-year-old twins in the same daycare for \$2650 per month. The Richards family spends \$3000 per month for their 3 year-old and infant twins to attend. Determine the monthly enrollment fee for the 4-year-old class, the 3-year-old class, and infant class.

4 yr old - x
3 yr old - y
infant - z

$$x + y + z = 2800$$

$$x + 2y = 2650$$

$$y + 2z = 3000$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2800 \\ 1 & 2 & 0 & 2650 \\ 0 & 1 & 2 & 3000 \end{array} \right] = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 850 \\ 0 & 1 & 0 & 900 \\ 0 & 0 & 1 & 1050 \end{array} \right]$$

4 yr olds are \$850

3 yr olds are \$900

and infants are \$1050.

8. A manufacturer of insulated cups has three machines that, when all working, produce the metal shell for 1225 cups per day. When only machine A and machine B are operational, they produce 675 shells per day. When only machine B and machine C are operational, they produce 1000 shells per day. Determine the daily quantity of metal shells produced by each machine.

$$A + B = 675$$

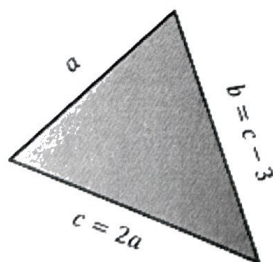
$$B + C = 1000$$

$$A + B + C = 1225$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 675 \\ 0 & 1 & 1 & 1000 \\ 1 & 1 & 1 & 1225 \end{array} \right] = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 225 \\ 0 & 1 & 0 & 450 \\ 0 & 0 & 1 & 550 \end{array} \right]$$

A produces 225, B produces 450 and C produces 550.

9. A triangle has a perimeter of 77 inches. What are the side lengths of a , b , and c ?



$$a + b + c = 77$$

$$b = c - 3$$

$$c = 2a$$

$$a + b + c = 77$$

$$b - c = -3$$

$$-2a + c = 0$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 77 \\ 0 & 1 & -1 & -3 \\ -2 & 0 & 1 & 0 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 14 \\ 0 & 1 & 0 & 29 \\ 0 & 0 & 1 & 32 \end{array} \right]$$

Side $a = 14$ in.
Side $b = 29$ in.
Side $c = 32$ in.

One family enrolled their two young children and teenager in summer camp for a total of \$900. Another family enrolled one young child, one older child, and one teenager for a total of \$1000. A third family enrolled one older child and two teenagers for a total of \$1450. Determine the cost of summer camp for each age group.

young child - x
 teenager - y
 older child - z

$$2x + y = 900$$

$$x + y + z = 1000$$

$$2y + z = 1450$$

$$\begin{bmatrix} 2 & 1 & 0 & | & 900 \\ 1 & 1 & 1 & | & 1000 \\ 0 & 2 & 1 & | & 1450 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & | & 150 \\ 0 & 1 & 0 & | & 600 \\ 0 & 0 & 1 & | & 250 \end{bmatrix}$$

Young \$150, teenager \$600, older child \$250

11. An assembly plant can assemble 20 cars per day when they run all three shifts. If only shift A and B work, the plant assembles 15 cars per day. If only shift B and C work, the plant assembles 13 cars per day. Determine the daily quantity of cars produced by each shift.

$$A + B + C = 20$$

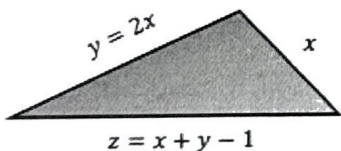
$$A + B = 15$$

$$B + C = 13$$

$$\begin{bmatrix} 1 & 1 & 1 & | & 20 \\ 1 & 1 & 0 & | & 15 \\ 0 & 1 & 1 & | & 13 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & | & 7 \\ 0 & 1 & 0 & | & 8 \\ 0 & 0 & 1 & | & 5 \end{bmatrix}$$

Shift A produces 7, Shift B 8, and Shift C 5.

12. A triangle has a perimeter of 11 feet. What are the side lengths of x , y , and z ?



$$x + y + z = 11$$

$$y = 2x$$

$$z = x + y - 1$$

$$x + y + z = 11$$

$$-2x + y = 0$$

$$-x - y + z = -1$$

$$\begin{bmatrix} 1 & 1 & 1 & | & 11 \\ -2 & 1 & 0 & | & 0 \\ -1 & -1 & 1 & | & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & | & 2 \\ 0 & 1 & 0 & | & 4 \\ 0 & 0 & 1 & | & 5 \end{bmatrix}$$

$x = 2$ ft. $y = 4$ ft. $z = 5$ ft.

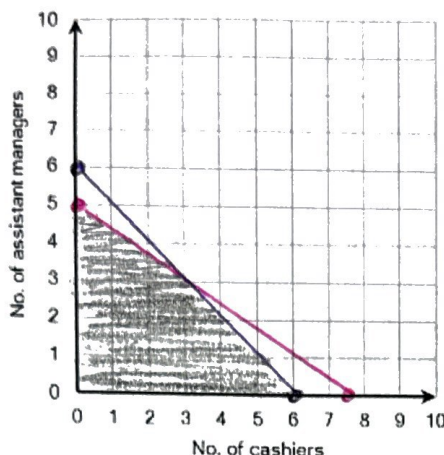
13.

Problem 4 A gift shop employs cashiers and assistant managers. Cashiers earn \$200 per week and assistant managers earn \$300 per week. The shop employs at most 6 employees at a given time. The gift shop's weekly payroll for cashiers and assistant managers cannot exceed \$1,500.

A. Write a system inequalities to represent the possible employment combinations.

$$\begin{aligned} c + m &\leq 6 \\ 200c + 300m &\leq 1500 \end{aligned}$$

B. Represent the system of inequalities on a graph.



$$c + m \leq 6$$

$$0 + m \leq 6$$

$$(0, 6)$$

$$c + 0 \leq 6$$

$$c \leq 6$$

$$(6, 0)$$

$$200(0) + 300m \leq 1500$$

$$300m \leq 1500$$

$$m \leq 5$$

$$(0, 5)$$

$$200c + 300(0) \leq 1500$$

$$200c \leq 1500$$

$$c \leq 7.5$$

$$(7.5, 0)$$

C. Prepare two possible employment scenarios (solutions) that satisfy the constraints of this situation.

Possible option 1: 2 cashiers and 3 assistant managers

Possible option 2: 4 cashiers and 2 "

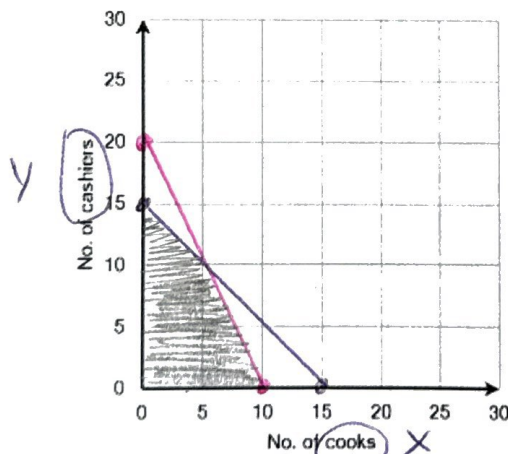
14.

Problem 4 A fast food restaurant employs cooks and cashiers. Cooks earn \$500 per week and cashiers earn \$250 per week. The restaurant employs at most 15 employees at a given time. The restaurant's weekly payroll for cooks and cashiers cannot exceed \$5,000.

A. Write a system inequalities to represent the possible employment combinations.

$$\begin{aligned} x + y &\leq 15 \\ 250y + 500x &\leq 5000 \end{aligned}$$

B. Represent the system of inequalities on a graph.



$$x + y \leq 15$$

$$0 + y \leq 15$$

$$(0, 15)$$

$$x + 0 \leq 15$$

$$(15, 0)$$

$$500x + 250y \leq 5000$$

$$500(0) + 250y \leq 5000$$

$$y \leq 20$$

$$(0, 20)$$

$$500x + 250(0) \leq 5000$$

$$500x \leq 5000$$

$$x \leq 10$$

$$(10, 0)$$

C. Prepare two possible employment scenarios (solutions) that satisfy the constraints of this situation.

possible option 1: 5 cooks, 5 cashiers

possible option 2: 4 cooks, 6 cashiers